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1- INTRODUCTION

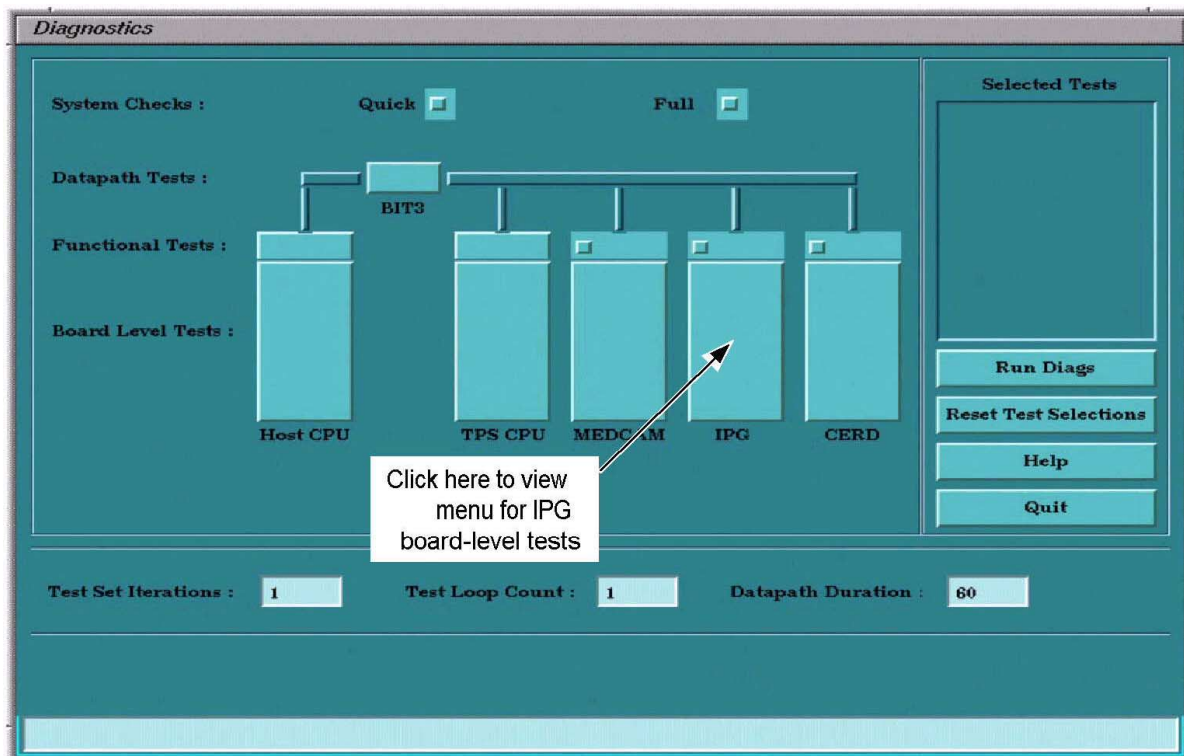
The TYME-DAC test sends basic waveforms, generated locally from the SPU, out from the IPG serial port through the TYME-II board D/A converter. The scope is connected to J6 (Auxiliary Analog Outputs) pin 1, 2, 3, and 4 to view the waveforms. The waveforms consist of a linear ramping pattern (sawtooth) of different frequencies. Unexpected rugged steps and spikes indicate TYME-DAC malfunction.

Note

Do not start multiple copies of diagnostics. Multiple copies of diagnostics have caused different anomalies such as locked up diagnostic windows.

2- PROCEDURE

1. From the Service Desktop, select **[Diagnostics]**, then **[Start...]**.
2. On the diagnostics menu click **[IPG]** to display the IPG board-level diagnostics menu (see Illustration 2-1.).



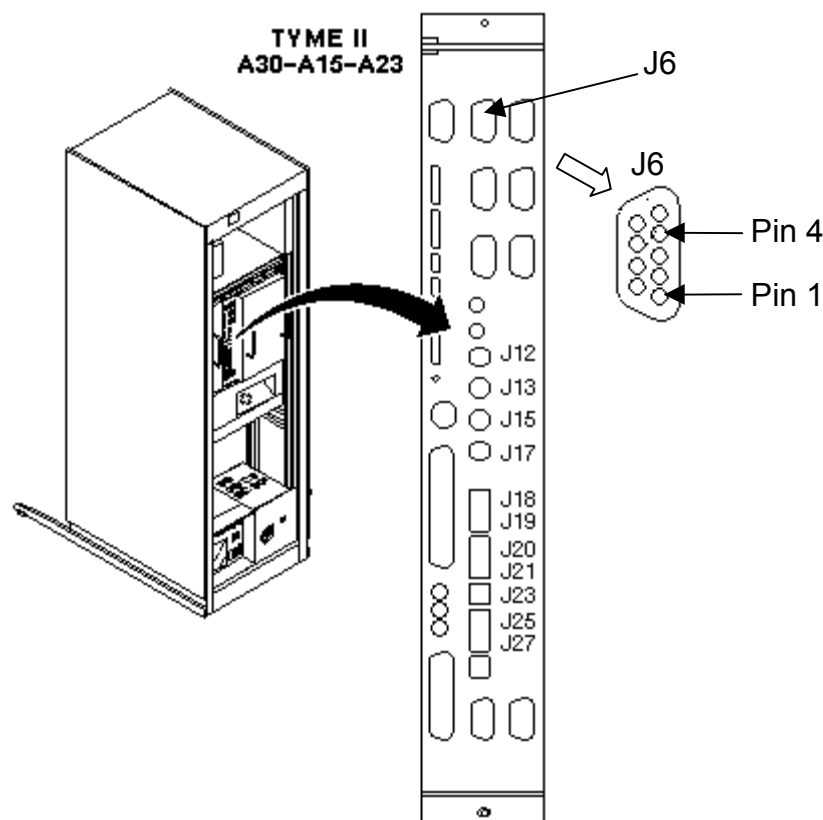
OPENING THE IPG BOARD-LEVEL TEST MENU
ILLUSTRATION 2-1

3. When the menu displays, click **[Manual...]**, then **[SPU TYME-II DAC]**. The following message is displayed at the bottom of the MAN window:
Disconnect cable from connector J6 on the TYME-II card. Verify sawtooth waveforms at pins 1, 2, 3, and 4 using an oscilloscope.

Note

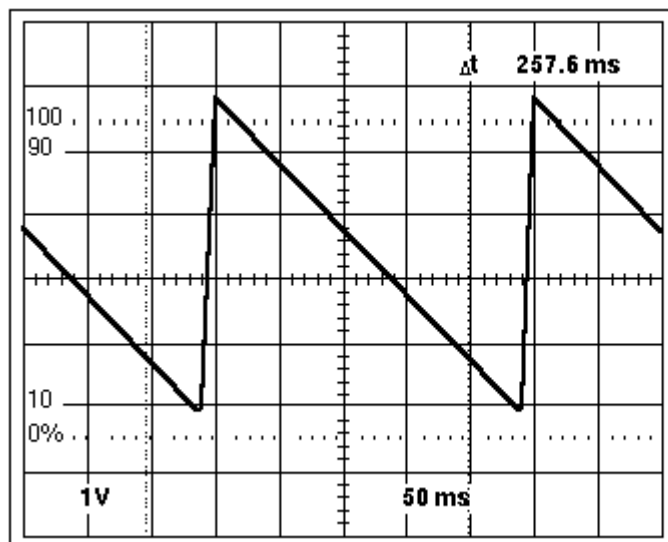
If no cable is connected to on J6 on the TYME-II card, disregard that part of the message.

4. Click **[Run Diags]**. After the TPS/ISE resets, the IPG Manual Diags will start.
5. Connect the oscilloscope ground to a suitable grounding point, and the probe to pins 1, 2, 3, and 4 of J6 as instructed. See Illustration 2-2.



TYME-II BOARD J6
ILLUSTRATION 2-2

6. Verify sawtooth waveforms per appropriate Illustration:
 - J6, Pin 1 - Illustration 2-3
 - J6, Pin 2 - Illustration 2-4
 - J6, Pin 3 - Illustration 2-5
 - J6, Pin 4 - Illustration 2-6



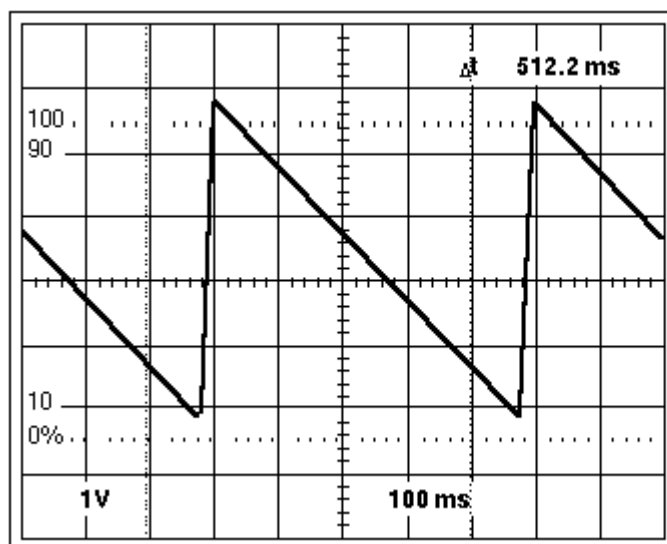
J6 Pin-1

Scope Channel: 1
50 ms/div
1V/div

Trigger:
Auto/Normal

L1647B

WAVEFORM AT J6, PIN 1
ILLUSTRATION 2-3



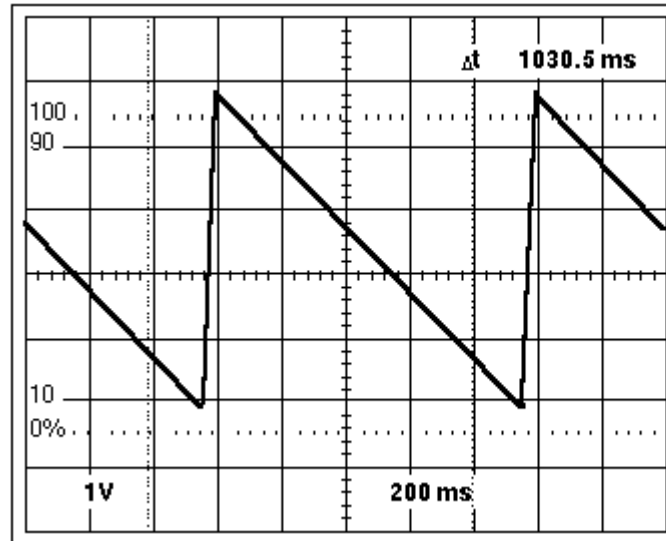
J6 Pin-2

Scope Channel: 1
100 ms/div
1V/div

Trigger:
Auto/Normal

L1648B

WAVEFORM AT J6, PIN 2
ILLUSTRATION 2-4



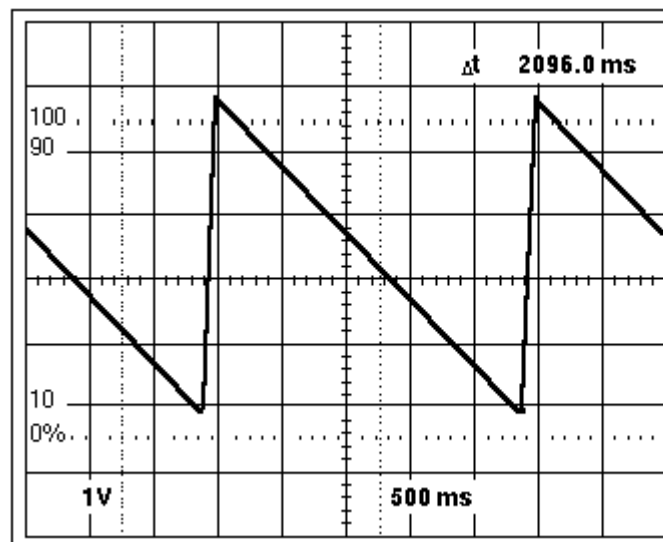
J6 Pin-3

Scope Channel: 1
200 ms/div
1V/div

Trigger:
Auto/Normal

L1649 B

WAVEFORM AT J6, PIN 3
ILLUSTRATION 2-5



J6 Pin-4

Scope Channel: 1
500 ms/div
1V/div

Trigger:
Auto/Normal

L1650 B

WAVEFORM AT J6, PIN 4
ILLUSTRATION 2-6

Note

These are very low frequency waveforms. Note the timebase setting.

3- SYSTEM RESTORATION

1. Disconnect oscilloscope probe.
2. Click **[Stop Diags]**, **[Close]**, **[Quit]**. The TPS/ISE will reset.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 29, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Nov. 10, 1999	M. Keber	Updated procedure for correct TMYE-II Board illustrations and connector number.
2	Oct. 31, 2000	M. Jones	Added illustration and text in Section 2- to clarify menu selection.